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Commutative algebra

Module *M03*

Avant-propos

This course workbook (version $\frac{1}{2}$ -course) is intended for students of "Master Mathematics and applications" at higher Normal School-Fez. It offers an incomplete course (without proofs) on commutative algebra: Rings and morphisms, Modules, Localization, tensor product, Chains condition, Integral extension, Krull dimension.

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Chapter 1

Rings and morphisms

1.1 Rings and morphisms

Definition 1.1.

A ring is a set with two binary operations $(R, +, \times)$ such that:

- $(R, +)$ is an Abelian group.
- The multiplication $\times$ is associative, that is, for all $a, b, c \in R$, $a(bc) = (ab)c$.
- The multiplication $\times$ is distributive with respect to the addition $+$, that is, for all $a, b, c \in A$, $(a + b) \times c = a \times c + b \times c$ et $a \times (b + c) = a \times b + a \times c$.

The ring R unitary if, $\times$ has an identity element called the unit, and denote it by 1_R or 1 .

The ring R is commutative if the multiplication $\times$ is commutative.

Throughout this course by **ring** we mean a commutative ring with identity.

Let $(A_i)_{i \in I}$ be a family of rings. Define two binary operations on the Cartesian product $\prod_{i \in I} A_i$ as follow:

$$(x_i)_{i \in I} + (y_i)_{i \in I} = (x_i + y_i)_{i \in I}, \quad (x_i)_{i \in I} (y_i)_{i \in I} = (x_i y_i)_{i \in I}$$

Then $(\prod_{i \in I} A_i, +, \times)$ is a commutative ring called the direct product of the rings $A_i, i \in I$.

A ring is trivial if it is a singleton, in this case $1 = 0$.

Let R be a ring and $a \in R$. We say that a is invertible if there exists $b \in R$ such that $ab = 1$, in this case b is unique and it is denoted a^{-1} . The set of invertible

elements of R is denoted $R^\times$ called the group of units. It is easy to see that $R^\times$ is a group (with respect to multiplication).

Definition 1.2.

Let R be a ring and $a \in R$.

1. a is a zero-divisor if there exists a non zero element $b \in R$ such that $ab = 0$. The set of zero-divisor is denoted $Z(R)$.
2. a is a nilpotent element if there exists $n \in \mathbb{N}$ such that $a^n = 0$.

Remark :

1. An invertible element is not a zero-divisor.
2. If R is a non trivial ring, any nilpotent element is a zero-divisor.
3. A zero divisor element is not necessarily a nilpotent element: $(1, 0)$ is a zero divisor in the ring $\mathbb{Z} \times \mathbb{Z}$, but it is not nilpotent.

Definition 1.3.

A ring R is an integral domain if;

1. R non trivial ($1 \neq 0$),
2. For all $a, b \in R$, $ab = 0 \implies a = 0$ or $b = 0$.

A ring R is an integral domain if R is non trivial and $Z(R) = \{0\}$.

Example :

1. $\mathbb{Z}$ is an integral domain.
2. $\mathbb{R}[X]$ is an integral domain.
3. $\mathbb{Z}^2$ is not integral, since $(1, 0)(0, 1) = 0$.

Definition 1.4.

A ring R is a field if

1. R is non trivial ($1 \neq 0$),

2. $R^\times = R \setminus \{0\}$, that is every nonzero element is invertible.

If R is a field then R is an integral domain.

Definition 1.5.

Let R be a ring and B be a subset of R . We say that B is a subring of R if, $(B, +)$ is a subgroup of $(R, +)$, B is stable with respect to multiplication and $1 \in B$.

Definition 1.6.

Let R and R' be rings and $f : A \rightarrow B$. We say that f is a morphism of rings if: for all $a, b \in A$, $f(a + b) = f(a) + f(b)$, $f(ab) = f(a)f(b)$ and $f(1) = 1$.

If $f : R \rightarrow R'$ is a rings morphism then $f(0) = 0$, $f(a - b) = f(a) - f(b)$, moreover if a is an invertible element of R then $f(a)$ is invertible and $(f(a))^{-1} = f(a^{-1})$.
Let $f : R \rightarrow R'$ be a morphism of rings. We say that f is an isomorphism if it is bijective.

Two rings R and R' are isomorphic if there is an isomorphism between them.

1.2 Ideals

Definition 2.1.

Let R be a ring and I be a subset of R . We say that I is an ideal of R if;

1. I is a subgroup of $(R, +)$,
2. $\forall a \in R, \forall x \in I, ax \in I$.

Remark :

1. $\{0\}$ et R are ideals of R .
2. If I is an ideal of R containing an invertible element, then $I = R$.

3. A non trivial ring R is a field if and only if $\{0\}$ and R are the only ideals of R .

Proposition 2.2.

Let R be a ring.

- If I and J are ideals of R , then $I + J := \{x + y \mid (x, y) \in I \times J\}$ is an ideal of R called the sum of I and J .
- If $(I_\alpha)_{\alpha \in \Gamma}$ is a family of ideals of R , then $\cap_{\alpha \in \Gamma} I_\alpha$ is an ideal of R .

Proof :
Let S be a subset of R . The intersection of all ideals of R containing S is an ideal of R , and is called the ideal generated by S it is denoted (S) . That is

$$(S) = \bigcap_{I \text{ ideal of } R, I \supseteq S} I$$

Proposition 2.3.

Let R be a ring and S be a nonempty subset of R , then

- $(S) = \left\{ \sum_{i=1}^m a_i s_i \mid m \geq 1, a_i \in R, s_i \in S \right\}$.
- $(\emptyset) = \{0\}$.

Proof :
Remark : If $S = \{s_1, \dots, s_n\}$, then $(S) = \left\{ \sum_{i=1}^n a_i s_i \mid a_i \in R \right\}$, also it is denoted $(s_1, \dots, s_n)$.

Definition 2.4.

Let R be a ring.

1. Let I, J be ideals of R . The product of I and J denoted IJ is the ideal generated by all elements ab , where $a \in I$ and $b \in J$.
2. Let $(I_\alpha)_{\alpha \in \Gamma}$ be a family of ideals of R . The sum of the ideals I_α , $\alpha \in \Gamma$ is the ideal generated by $\bigcup_{\alpha \in \Gamma} I_\alpha$ it is denoted $\sum_{\alpha \in \Gamma} I_\alpha$

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The elements of IJ are $\sum_{k=1}^m a_k b_k$, where $a_k \in I$ and $b_k \in J$.

The elements of $\sum_{\alpha \in \Gamma} I_\alpha$ are $\sum_{\alpha \in \Gamma} a_\alpha$, where $a_\alpha \in I_\alpha$ and all but finite set are zero.

The product of a finitely many ideals $I_1, \dots, I_m$, is the ideal generated by all products $a_1 \cdots a_m$, where $a_k \in I_k$.

Definition 2.5.

Let R be a ring and I be an ideal of R .

1. We say that I is a principal ideal if it is generated by one element, that is $I = (a)$, where $a \in R$.
2. We say that I is a finitely generated ideal if it is generated by a finitely many elements, that is $I = (a_1, \dots, a_n)$ where $a_1, \dots, a_n \in R$.

Example :

1. In the ring $\mathbb{Z}[X]$, the ideal $I = \{P \in \mathbb{Z}[X] \mid P(0) = 0\}$ is principal. In fact $I = (X)$.
2. If I and J are finitely generated ideals, then so are IJ and $I + J$. In fact; ...

Definition 2.6.

A ring is a principal domain¹ if it is an integral domain and all its ideals are principal.

Examples :

1. $\mathbb{Z}$ is a principal domain.
2. $\mathbb{R}[X]$ is a principal domain.

Let I be an ideal of R . The quotient group R/I inherits a uniquely defined multiplication from R ($\bar{x} \bar{y} = \overline{xy}$), which makes it a (commutative) ring, called the quotient ring. Moreover, the map $\pi : R \rightarrow R/I$ defined by $\pi(x) = \bar{x}$ is a rings morphism.

¹It is quasi-principal if its ideals are principal (without integral condition).

Proposition 2.7.

Let R be a ring and I be an ideal of R . The morphism $\pi : A \rightarrow A/I$ induce a bijection between ideal of R/I and ideals of R containing I .

Proof :

Proposition 2.8.

Let $f : R \rightarrow R'$ be a rings morphism.

1. If J is an ideal of R' then $f^{-1}(J)$ is an ideal of R . In particular $\ker f$ is an ideal of R .
2. $\text{Im } f$ is a subring of R' .

Proof :

Theorem 2.9. (First isomorphism theorem)

Let $f : R \rightarrow R'$ be rings. Then the map

$$\bar{f} : A/\ker f \rightarrow \text{Im } f, \bar{f}(\bar{x}) = f(x)$$

is well defined and is an isomorphism.

Proof :

Remarks :

1. Second isomorphism theorem: Let R' be a subring of a ring R and I be an ideal of R . Then $R' + I$ is a subring of R , $R' \cap I$ is an ideal of R , moreover the rings $(R' + I)/I$ et $R'/(R' \cap I)$ are isomorphic.
2. Third isomorphism theorem: Let R be a ring and $I \subseteq J$ be ideals of R . Then J/I is an ideal of R/I , moreover the rings $(R/I)/(J/I)$ and R/J are isomorphic.

1.3 Prime and maximal ideals

Definition 3.1.

Let R be a ring.

1. An ideal P is a prime ideal of R if $P \neq R$ and $\forall x, y \in R, xy \in P \Rightarrow x \in P$ or $y \in P$.
2. An ideal M is a maximal ideal of R if $M \neq R$ and there is no ideal I of R such that $M \subset I \subset R$ (strict inclusion).

The set of prime ideals of R is called the prime spectrum of R and is denoted $\text{Spec}(R)$, whereas the set of maximal ideals of R is called maximal spectrum of R and is denoted $\text{MaxSpec}(R)$ or $\text{Max}(R)$.

Examples :

1. In the ring $\mathbb{R}[X]$ the ideal (X) is prime.
2. In the ring $\mathbb{Z}$ the ideal (3) is maximal.
3. In the ring $\mathbb{Z}^2$ the ideal $((0, 6))$ is not a prime ideal.

Proposition 3.2.

Let R be a ring.

1. An ideal P is prime if and only if R/P is an integral domain.
2. An ideal M is maximal if and only if R/M is a field.

Proof :

Proposition 3.3.

1. A maximal ideal is a prime ideal.
2. Let $f : R \rightarrow R'$ be a rings morphism. If P is a prime ideal of R' then $f^{-1}(P)$ is a prime ideal of R .

Proof :

Theorem 3.4. (Krull)

Let R be a non trivial ring i.e $R \neq 0$, then $\text{Max}(R) \neq \emptyset$. That is R contains at last a maximal ideal.

Proof : (by Zorn's Lemma)

Corollary 3.5.

Let R be a ring.

1. *Every proper ideal (i.e $\neq R$) of R is contained in a maximal ideal.*
2. *If $a \in R$ is non invertible, then a is contained in a maximal ideal.*

Proof :

Remark : As a consequence of the previous corollary;

$$A^\times = A \setminus \left(\bigcup_{M \in \text{Max}(A)} M \right)$$

Definition 3.6.

Let R be a ring and $P \in \text{Spec}(R)$. We say that P is a minimal prime ideal of R if it is minimal in the set $\text{Spec}(R)$ with respect to inclusion, that is if $Q \in \text{Spec}(R)$ such that $Q \subseteq P$ then $Q = P$.

Examples :

1. (0) is a minimal prime ideal of $\mathbb{Z}$. Moreover if R is an integral domain, then (0) is a minimal prime ideal of R .
2. $\mathbb{Z} \times \{0\}$ is a minimal prime ideal of $\mathbb{Z} \times \mathbb{Z}$.

Theorem 3.7.

Let R be a ring and $P \in \text{Spec}(R)$. Then P contains at last a minimal prime ideal.

Proof :

Remarks :

1. Every non trivial ring has at last a minimal prime ideal.
2. A minimal prime ideal can contain strictly an ideal.

Theorem 3.8. (Avoidance Lemma)

Let R be a ring, let $P_1, \dots, P_n \in \text{Spec}(R)$ and I be an ideal of R such that $I \subseteq \bigcup_{k=1}^n P_k$. Then $I \subseteq P_l$ for some $1 \leq l \leq n$.

Proof :

Theorem 3.9. (Intersection Lemme)

Let R be a ring, let $P \in \text{Spec}(R)$ and $I_1, \dots, I_n$ be ideals of R such that $\bigcap_{k=1}^n I_k \subseteq P$. Then there exists $1 \leq l \leq n$ such that $I_l \subseteq P$.

Proof :

1.4 Radical

An element $a \in A$ is nilpotent if there exists $n \in \mathbb{N}$ such that $a^n = 0$. The set of nilpotent elements of R is denoted $\text{Nil}(R)$ called the nilradical of R . That is

$$\text{Nil}(R) = \{a \in R \mid \exists n \in \mathbb{N}, a^n = 0\}$$

Proposition 4.1.

Let R be a ring. The nilradical of R is an ideal of R .

Proof :

Theorem 4.2.

Let R be a ring. Then

$$\text{Nil}(A) = \bigcap_{P \in \text{Spec}(A)} P$$

Proof :

Definition 4.3.

Let R be a ring and I be an ideal of R . The radical of I denoted $\sqrt{I}$ is;

$$\sqrt{I} := \{a \in R \mid \exists n \in \mathbb{N}, a^n \in I\}$$

Examples :

1. $\sqrt{(0)} = \text{Nil}(R)$.
2. In the ring $\mathbb{Z}$, $\sqrt{(8)} = (2)$, and $\sqrt{(24)} = (6)$.

Proposition 4.4.

Let R be a ring and I, J be ideals of R .

1. $\sqrt{I}$ is an ideal of R containing I .
2. If $I \subseteq J$, then $\sqrt{I} \subseteq \sqrt{J}$.
3. $\sqrt{\sqrt{I}} = \sqrt{I}$.
4. $\sqrt{IJ} = \sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J}$.
5. $\sqrt{I+J} = \sqrt{\sqrt{I} + \sqrt{J}}$.

Proof :

Theorem 4.5.

Let R be a ring and I be an ideal of R . Then

$$\sqrt{I} = \bigcap_{P \in \text{Spec}(R), I \subseteq P} P$$

Proof :

Definition 4.6.

Let R be a ring. The Jacobson radical of R denoted $J(R)$ is the ideal

$$J(R) := \bigcap_{M \in \text{Max}(R)} M$$

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Remark : Let R be a ring, then $\text{Nil}(R) \subseteq J(R)$.

Proposition 4.7.

Let R be a ring and $a \in R$. Then $a \in J(R)$ if and only if for all $x \in R$, $1 - ax \in R^\times$. That is

$$J(R) = \{a \in R \mid \forall x \in R, 1 - ax \in R^\times\}$$

Proof :

1.5 Exercises

Exercise 1.1

Let R be a ring. Let $(a, x) \in \text{Nil}(R) \times R^\times$. Show that $a + x \in R^\times$.

Exercise 1.2

Let R be a ring and I be an ideal of R such that $\bigcap_{n \geq 1} I^n = (0)$. Show that for any $x \in I$, $1 + x \notin Z(A)$.

Exercise 1.3

Let R be a ring, $(P_\alpha)_{\alpha \in \Gamma}$ be a family of prime ideals of R , and $I = \bigcap_{\alpha \in \Gamma} P_\alpha$. Show that $\sqrt{I} = I$.

Exercise 1.4

Let R be a ring such that for any $x \in R$, there exists $n \geq 2$ such that $x^n = x$. Show that every prime ideal is maximal.

Exercise 1.5

Let R be a ring such that every prime ideal is principal. We will show that every ideal is principal. Assume the contrary, that is R has a non

principal ideal. Set

$$X = \{I \text{ ideal of } R \mid I \text{ not principal}\}$$

1. Show that X is a nonempty inductive set.
2. Deduce that X has a maximal element, say P .
3. Let $x, y \in R$ such that $xy \in P$ and $x, y \notin P$. Set $J = P + (y)$ and $J' = (P : J) = \{z \in A \mid zJ \subseteq P\}$.
 - (a) Show that J and J' are principal ideals.
 - (b) Show that $P = JJ'$, deduce that P is principal.
4. Show that P is prime.
5. Conclusion.

Exercise 1.6

Let $R_1, \dots, R_m$ be rings, $R = R_1 \times \dots \times R_m$ their direct product. For $1 \leq k \leq m$, denote $s_k : R \rightarrow R_k$ the k th projection.

1. For $1 \leq k \leq m$, let I_k be an ideal of R_k . Show that $I_1 \times \dots \times I_m$ is an ideal of R .
2. Let I be an ideal of R . Set $I_k = s_k(I)$.
 - (a) For $1 \leq k \leq m$, show that I_k is an ideal of R_k .
 - (b) Show that $I = I_1 \times \dots \times I_m$.
3. Let $P \in \text{Spec}(A)$, write $P = P_1 \times \dots \times P_m$, where P_k is an ideal of R_k . For $1 \leq k \leq m$, denote e_k the element of R whose components are all zero except the k th, which is equal to 1.
 - (a) Show that $\sum_{k=1}^m e_k = 1 (= 1_A)$, $e_k e_l = 0$ if $k \neq l$ and $e_k^2 = e_k$.
 - (b) Show that there exists $1 \leq j \leq m$ such that $e_j \notin P$.
 - (c) Let $1 \leq k \leq m$ with $k \neq j$. Show that $e_k \in P$.

- (d) Deduce that $P_k = A_k$ si $k \neq j$ and that P_j is a prime ideal of R_j .
- (e) Show that if P is a maximal ideal of R then P_j is a maximal ideal of R_j .

Exercise 1.7

Let R be an integral domain. Show that $A[X]$ is an integral domain.

Exercise 1.8

Let R be a ring. Show that $A[X]$ is a principal domain if and only if R is a field.

Exercise 1.9

Let R be a ring and $P = \sum_{k=0}^n a_k X^k \in A[X]$.

1. Show that P is invertible if and only if, a_0 is invertible and $a_1, \dots, a_n$ are nilpotents.
2. Show that P is nilpotent if and only if the a_k are nilpotents.
3. Deduce that $\text{Nil}(A[X]) = J(A[X])$.

Exercise 1.10

Let R be a ring and I be an ideal of R . Set

$$I[X] = \left\{ \sum_{k=0}^n a_k X^k \in A[X] \mid n \geq 0, a_k \in I \right\}$$

$I[X]$ the set of polynomials with coefficients in I .

1. Show that $I[X]$ is an ideal of $R[X]$.
2. Show that $A[X]/I[X]$ is isomorphic to $(A/I)[X]$.

3. Deduce that I is prime if and only if $I[X]$ is prime.

Exercise 1.11

Let R be a ring and $I_1, \dots, I_m$ be ideals of R such that $\sum_{k=1}^m I_k = R$. Let $l_1, \dots, l_m \in \mathbb{N}$. Show that $\sum_{k=1}^m I_k^{l_k} = R$.

Exercise 1.12

Let R be a ring and I, J be ideals of R .

1. Show that if $\sqrt{IJ} = A$ then $I = J = R$.
2. Let $P \in \text{Spec}(R)$ such that $IJ = P$. Show that $I = P$ or $J = P$.

Exercise 1.13 (Zarisky Topology)

Let R be a ring. For I ideal of R , denote $V(I)$ the set of prime ideals of R containing I , that is

$$V(I) = \{P \in \text{Spec}(R) \mid I \subseteq P\}$$

1. Let I be an ideal of R . Show that $V(I) = V(\sqrt{I})$.
2. Let I, J be ideals of R . Show that $V(I) = V(J)$ if and only if $\sqrt{I} = \sqrt{J}$.
3. Show that $V(0) = \text{Spec}(A)$ and $V(A) = \emptyset$.
4. Let I, J be ideals of R . Show that $V(I \cap J) = V(I) \cup V(J)$.
5. Let $(I_\alpha)_{\alpha \in \Gamma}$ be a family of ideals of R . show that $\bigcap_{\alpha \in \Gamma} V(I_\alpha) = V(\sum_{\alpha \in \Gamma} I_\alpha)$.

We deduce, from the previous properties, that there exists a unique topology of $\text{Spec}(R)$ whose closed subsets are the $V(I)$,

where I is an ideal of R . this topology is called the Zariski Topology on $\text{Spec}(R)$. For I ideal of R denote $D(I) = \text{Spec}(A) \setminus V(I) = \{P \in \text{Spec}(A) \mid I \not\subseteq P\}$.

6. Show that $D((f))$, where $f \in A$ form a basis^a of open subsets with respect to Zariski topology.
7. Show that $\text{Spec}(A)$ is quasi-compact.
8. Let $f \in A$. Show that $D(f) = \emptyset$ if and only if $f \in \text{Nil}(A)$.
Let $f : A \rightarrow B$ be a morphism of rings. denote $f^* : \text{Spec}(B) \rightarrow \text{Spec}(A)$ the map defined for each $P \in \text{Spec}(B)$ by $f^*(P) = f^{-1}(P)$.
9. Let I be an ideal of R . Show that $f^{*-1}(V(I)) = V(f(I)B)$ (where $f(I)B$ is that ideal of B generated by $f(I)$).
10. Deduce that f^* is continuous.
11. Let I be an ideal of R . Show that $\text{Spec}(A/I)$ is homeomorphic to $V(I)$

^aThat is, stable under finite intersection and each open is union of elements of this basis

Exercise 1.14

Let R be a ring, we endowed $\text{Spec}(R)$ with the Zariski topology. Show that $\text{Spec}(A)$ est connected id and only if the only idempotent^a elements of R are 0 et 1.

^aAn element $a \in R$ is idempotent if $a^2 = a$

Exercise 1.15

Let I be an ideal of the ring R and $P \in \text{Spec}(R)$.

1. Show that $\overline{\{P\}} = V(P)$.
2. Deduce that $\{P\}$ is a closed subset if and only if $P \in \text{Max}(R)$.

Exercise 1.16

Let I be an ideal of R . Show that $D(I)$ is quasi-compact if and only if there exists $a_1, \dots, a_m \in R$ such that $\sqrt{I} = \sqrt{(a_1, \dots, a_m)}$.

Exercise 1.17

Let R be a ring, let I be an ideal of R and $a \in R$ such that $I + (a)$ and $(I : a) := \{x \in R \mid xa \in I\}$ are finitely generated.

1. Show that there exists $d_1, \dots, d_m \in I$ such that $I + (x) = (d_1, \dots, d_m, x)$.
2. Let x be an element of I of the form $x = \sum_{k=1}^m \alpha_k d_k + r x \in I$. Justifies that $r \in (P : a)$.
3. Deduce that I is finitely generated.

Exercise 1.18

Let R be a ring whose prime ideals are finitely generated. Set

$$X = \{I \text{ idéal de } R \mid I \text{ not finitely generated}\}$$

Assume that X is not empty.

1. Show that X has a maximal element P .
2. Show that there exists $a, b \in R$ such that $ab \in P$ and $a, b \notin P$.
3. Show that $P + (a)$ and $(P : a)$ are finitely generated.
4. Deduce that P is finitely generated.
5. Conclude.

Chapter 2

Modules

2.1 Modules

Definition 1.1.

Let R be a commutative ring. An R -module or module over R is a set M together with two operations $(M, +, \cdot)$ such that;

1. $(M, +)$ is an Abelian group.
2. $\cdot : R \times M \rightarrow M, (a, m) \mapsto a.m$ (scalar multiplication) such that; for all $a, b \in R, m, m' \in M$,
 - $a.(m + m') = a.m + a.m'$
 - $(a + b).m = a.m + b.m$
 - $(ab).m = a.(b.m)$
 - $1.m = m$

Remark : Modules are generalization of vector spaces, in the sense that; modules over commutative field are exactly vector spaces.

Examples :

1. Let G be an Abelian group. Then G is naturally a $\mathbb{Z}$ -module, where the scalar multiplication “ $\cdot$ ” is defined by $\mathbb{Z} \times G \rightarrow G, (n, g) \mapsto ng$.
2. If I is an ideal of a ring R , then I is naturally an R -module.

3. If I is an ideal of a ring R then R/I is an R -module.

Definition 1.2.

Let M and N be R -modules and $f : M \rightarrow N$ be a map. We say that f is a morphism of R -modules if, for all $a \in R$, $m, m' \in M$;

1. $f(m + m') = f(m) + f(m')$.
2. $f(am) = af(m)$.

Examples :

1. Let I be an ideal of a ring R . The map $s : R \rightarrow R/I$, $x \mapsto s(x) = \bar{x}$ is a morphism of R -modules.
2. A groups morphism is a morphism of $\mathbb{Z}$ -modules.

A composition of morphisms of R -modules is a morphism of R -modules.

The set of all morphisms of R -modules from M to N is an R -module, where for f, g morphisms of R -modules and $a \in R$, $(f + g)(x) = f(x) + g(x)$ and $(af)(x) = af(x)$. We denote $\text{Hom}_A(M, N)$ or simply $\text{Hom}(M, N)$ the R -module of all morphisms of R -modules from M to N .

Definition 1.3. (R -algebra)

Let R be a ring. An R -algebra is quadruple $(B, +, \times, .)$ such that ;

1. $(B, +, \times)$ is a ring,
2. $(B, +, .)$ is an R -module,
3. $a(xy) = (ax)y = x(ay)$, $\forall a \in A, x, y \in B$.

Examples : Let R be a ring.

1. R is naturally a $\mathbb{Z}$ -algebra.
2. $R[X]$ is an R -algebra.
3. Let $f : R \rightarrow R'$ be a morphism of rings. Then R' is an R -module ($a.b := f(a)b$), moreover it is an R -algebra.

2.2 Submodules

Definition 2.1.

Let M be an R -module et N be a subset of M . We say that N is a submodule of M if;

1. N is a subgroup of $(M, +)$,
2. $\forall a \in R, \forall n \in N, an \in N$.

Remark : Si N est un sous module de M , alors N lui même est un A -module.

Examples :

1. Si I est un idéal de A , alors I est un sous module de A .
2. Soit $f : M \rightarrow N$ un morphisme de A -modules. Alors $\ker f$ est un sous module de M et $\text{Im } f$ est un sous module de N .
3. Si $x \in M$, alors $\{ax / a \in A\}$ est un sous module de M et il se note Ax .

Proposition 2.2.

Let M be an R -module.

1. If $(N_i)_{i \in I}$ is a family of submodules of M , then $\cap_{i \in I} N_i$ is a submodule of M .
2. If N and N' are submodules of M , then $N + N'$ is a submodule of M , where $N + N' = \{x + y / x \in N, y \in N'\}$.

Proof :

1. Observe that $\cap_{i \in I} N_i$ is a subgroup of M , since it is an intersection of subgroups of M . Let $a \in R$ and $x \in \cap_{i \in I} N_i$. then, for all $i \in I$, $x \in N_i$, so for all $i \in I$, $ax \in N_i$. Hence $ax \in \cap_{i \in I} N_i$.
2. Note that $0 = 0 + 0 \in N + N'$. If $z = x + y$ et $z' = x' + y'$ are in $N + N'$ where $x, x' \in N$ and $y, y' \in N'$, then $z - z' = (x - x') + (y - y') \in N + N'$, that is $N + N'$ is a subgroup of M . Moreover, if $a \in R$, then $az = ax + ay \in N + N'$.

2.3 Submodules generated by a family

Let M be an R -module and N be a submodule of M . Then M/N is an Abelian group, where the addition is defined by;

$$\bar{x} + \bar{y} = \overline{x + y}, \quad x, y \in M$$

It is also an R -module, where the scalar multiplication is defined by;

$$a\bar{x} = \overline{ax}, \quad a \in R, x \in M.$$

Moreover $s : M \rightarrow M/N$ defined by $s(x) = \bar{x}$ is a (surjective) morphism of R -modules.

Proposition 2.3.

Let M be an R -module and N be a submodule of M . If $f : M \rightarrow M'$ is a morphism of R -modules such that $N \subseteq \ker f$, then there exists a unique morphism of R -modules $\bar{f} : M/N \rightarrow M'$ such that $\bar{f} = f \circ s$ where $s : M \rightarrow M/N$ such that $s(x) = \bar{x}$.

$$\begin{array}{ccc} M & \xrightarrow{f} & M' \\ s \downarrow & \nearrow \bar{f} & \\ M/N & & \end{array}$$

Proof :

Theorem 2.4. (isomorphism theorem)

Let $f : M \rightarrow M'$ be a morphism of R -modules. Then f induce an isomorphism $\bar{f} : M/N \rightarrow \text{Im } f$ where $\bar{f}(\bar{x}) = f(x)$.

Proof :

2.3 Submodules generated by a family

Definition 3.1.

Let M be an R -module and $(x_i)_{i \in I}$ be a family of elements of M . The submodule generated by the family $(x_i)_{i \in I}$ is the intersection of all submodules of M containing the elements $x_i, i \in I$, It is denoted $(x_i, i \in I)$.

It is the smallest submodule of M containing the elements of the family $(x_i)_{i \in I}$.

Proposition 3.2.

Let M be an R -module and $(x_i)_{i \in I}$ be a family of elements of M . Then

$$(x_i, i \in I) = \left\{ \sum_{j \in J} a_j x_j \mid J \text{ finite subset of } I, a_j \in R \right\}$$

Proof :

Examples : Let M be an R -module.

1. Let $x \in M$. The submodule generated by x is $Rx = \{ax \mid a \in R\}$.
2. If $x_1, \dots, x_n \in M$, then $(x_1, \dots, x_n) = \{a_1 x_1 + \dots + a_n x_n \mid a_i \in R, \text{ for all } 1 \leq i \leq n\}$.
3. Let I be an ideal of R and N be a submodule of M . We denote IN the submodule of M generated by all elements ax where $a \in I$ and $x \in N$, then

$$IN = \left\{ \sum_{i=1}^n a_i x_i \mid n \in \mathbb{N}^*, a_i \in I, x_i \in N \right\}$$

Remark : Note that if $(N_i)_{i \in I}$ is a family of submodules of M , then $\cup_{i \in I} N_i$ is not in general a submodule of M .

Definition 3.3.

Let M be an R -module and $(N_i)_{i \in I}$ be a family of submodules of M . The sum of the submodules $N_i, i \in I$ is the submodule generated by $\cup_{i \in I} N_i$, it is denoted $\sum_{i \in I} N_i$. That is

$$\sum_{i \in I} N_i = (\cup_{i \in I} N_i)$$

Proposition 3.4.

Let M be an R -module and $(N_i)_{i \in I}$ be a family of submodules of M .
Then

$$\sum_{i \in I} N_i = \left\{ \sum_{j \in J} x_j \mid J \text{ finite subset of } I, x_j \in N_j \right\}$$

Proof :

Remarks : Let M be an R -module.

1. If $N_1, \dots, N_n$ are submodules of M , then

$$\sum_{i=1}^n N_i = \left\{ \sum_{i=1}^n x_i \mid \text{pour } 1 \leq i \leq n, x_i \in N_i \right\}$$

2. If $(x_i)_{i \in I}$ is a family of elements of M , then $\sum_{i \in I} Rx_i$ is the submodule generated by $(x_i)_{i \in I}$, that is

$$\sum_{i \in I} Rx_i = (x_i, i \in I)$$

Definition 3.5.

Let M be an R -module. We say that M finitely generated if there exists $m_1, \dots, m_r \in M$ such that $M = (m_1, \dots, m_r)$, that is M is generated by finitely many elements.

Proposition 3.6.

Let M be an R -module.

1. If M is finitely generated and N is a submodule of M , then M/N is finitely generated.
2. M is finitely generated if and only if M is (isomorphic to) a quotient of A^n for some $n \in \mathbb{N}^*$.

Proof :

2.4 Product and direct sum of modules

Let $(M_i)_{i \in I}$ be a family of R -modules. The product $\prod_{i \in I} M_i$ is an R -module where the addition and scalar multiplication are defined by;

$$(x_i)_{i \in I} + (y_i)_{i \in I} = (x_i + y_i)_{i \in I} \quad \text{et} \quad a(x_i)_{i \in I} = (ax_i)_{i \in I}$$

Called the direct product of R -modules M_i . The direct sum of R -modules M_i , denoted $\bigoplus_{i \in I} M_i$ is defined as follow

$$\bigoplus_{i \in I} M_i := \{(x_i)_{i \in I} \in \prod_{i \in I} M_i / \text{all but finitely } x_i \text{ are zero} \}$$

Remarks : Let $(M_i)_{i \in I}$ be a family of R -modules.

1. The direct sum $\bigoplus_{i \in I} M_i$ is isomorphic to a submodule of $\prod_{i \in I} M_i$.
2. If I is finite, then $\bigoplus_{i \in I} M_i$ and $\prod_{i \in I} M_i$ are isomorphic.

For $j \in I$, denote $\pi_j : \prod_{i \in I} M_i \rightarrow M_j$ the j th projection and $s_j : M_j \rightarrow \bigoplus_{i \in I} M_i$ the j th injection.

Proposition 4.1.

Let $(M_i)_{i \in I}$ be a family of R -modules and M be an R -module.

1. If for each $i \in I$, $f_i : M_i \rightarrow M$ is a morphism, then there exists a unique morphism $f : \bigoplus_{i \in I} M_i \rightarrow M$ such that, for each $j \in I$, the following diagram is commutative;

$$\begin{array}{ccc} M_j & \xrightarrow{f_j} & M \\ s_j \downarrow & \searrow f & \\ \bigoplus_{i \in I} M_i & & \end{array}$$

2. If for each $i \in I$, $g_i : M \rightarrow M_i$ is a morphism, then there exists a unique morphism $g : M \rightarrow \prod_{i \in I} M_i$ such that for each $j \in I$, the following diagram is commutative;

$$\begin{array}{ccc} M & \xrightarrow{g_j} & M_j \\ \searrow g & & \uparrow \pi_j \\ & & \prod_{i \in I} M_i \end{array}$$

Proof :
 Let I be any set, for each $i \in I$, set $A_i = A$ then the direct sum $\bigoplus_{i \in I} A_i$ denoted $\bigoplus_{i \in I} A$ or $A^{(I)}$. If $I = \{i_1, \dots, i_n\}$ is finite, in this case we have $R^{(I)} = R \oplus \dots \oplus R = R \times \dots \times R = R^n$.
 If $(M_i)_{i \in I}$ is a family of R -modules that are all isomorphic to R , then $\bigoplus_{i \in I} M_i$ is isomorphic to $A^{(I)}$.

Definition 4.2.

Let M be an R -module. We say that M is free if it is isomorphic to an R -module of the form $A^{(I)}$ where I is a set.

Examples :

1. The R -module $R[X]$ is free, since it is isomorphic to $A^{(\mathbb{N})}$.
2. The R -module R^n where $n \in \mathbb{N}^*$ is finitely generated free module.

Proposition 4.3.

Let R be a ring and M be an R -module. Then M is finitely generated if and only if there exists $n \in \mathbb{N}^$ such that M is isomorphic to a quotient of A^n .*

Proof :

Proposition 4.4.

Let R be a ring and $m, n \in \mathbb{N}$. If R^n and R^m are isomorphic then $n = m$.

Proof :

Theorem 4.5.

Let R be a ring and M be an R -module. If $(e_1, \dots, e_n)$ and $(\varepsilon_1, \dots, \varepsilon_m)$ are basis of M , then $n = m$.

Proof :

Definition 4.6.

We say that M is a finitely generated free module of finite rank if M has a finite basis. Its rank denoted $\text{rank} M$ is the number of element any basis.

Remark : M is finitely generated of rank n if and only if it is isomorphic to R^n .

2.5 Exercises

Exercise 2.1

Let N be a submodule of a module M over a ring R . Show that:

1. If N and M/N are finitely generated, then so is M .
2. If M is finitely generated, then so is M/N .

Exercise 2.2

Let $R = (\mathbb{Z}/2\mathbb{Z})^{\mathbb{N}}$ the ring of all sequences with coefficients in $\mathbb{Z}/2\mathbb{Z}$. Let $I = \{(u_n)_{n \in \mathbb{N}} / \exists N \in \mathbb{N}, u_n = 0, \forall n \geq N\}$.

1. Show that I is an ideal of R (so a submodule of R).
2. Prove that I is not finitely generated.

So submodule of finitely generated module need not be finitely generated.

Exercise 2.3

Let M be an R -module. Show that the following statements are equivalent:

1. The module M has only the trivial submodules 0 and M .
2. M is isomorphic to R/P for a maximal ideal P .

Exercise 2.4 (Nakayama lemma)

Let M be a finitely generated R -module and I be an ideal of R such that $I \subseteq J(R)$ and $IM = M$. We will to show that $M = 0$. Suppose that $M \neq 0$. Let

$$X = \{N \mid N \text{ submodule of } M \text{ and } N \neq M\}$$

1. Show that X has a maximal element L .
Let $x \in M \setminus L$.
2. Prove that $L + (x) = M$.
Set $J = \{a \in R \mid ax \in L\}$.
3. Show that J is a maximal ideal of R , deduce that $I \subseteq J$.
4. Show that $M = L$.
5. Conclude.

Exercise 2.5

Let R be a ring and M be an R -module. Consider the map $\varphi : \text{Hom}(R, M) \rightarrow M$ defined by $\varphi(f) = f(1)$. Prove that φ is an isomorphism, and describe its inverse.

Exercise 2.6

For N, N' submodules of an R -module M , define

$$(N : N') = \{a \in R \mid aN' \subseteq N\} = \{a \in R \mid \forall n \in N', an \in N\}$$

1. Prove that $(N : N')$ is an ideal of R .
2. Prove that $(N : N')N' \subseteq N$.
3. Show that $(N : N') = R$ if and only if $N' \subseteq N$.

A sequence of R -modules and morphisms of R -modules

$$\cdots \xrightarrow{f_{i-2}} M_{i-1} \xrightarrow{f_{i-1}} M_i \xrightarrow{f_i} M_{i+1} \xrightarrow{f_{i+1}} \cdots$$

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is called exacte at i if $\text{Im } f_{i-1} = \ker f_i$. It is called exacte if it is exacte at every i .

Exercise 2.7

Let M be an R -module and N a submodule of N . Show that the sequence $0 \longrightarrow N \xrightarrow{i} M \xrightarrow{s} M/N \longrightarrow 0$ is exacte (where $i(x) = x$ and $s(x) = \bar{x}$).

Exercise 2.8

Let $0 \longrightarrow M_0 \xrightarrow{f} M_1 \xrightarrow{g} M_2 \longrightarrow 0$ be a exacte sequence of R -modules.

1. Show that f is injective and g is surjective.
2. Prove that, if M_1 is finitely generated then so is M_2 .
3. Prove that, if M_0 and M_2 are finitely generated then so is M_1 .

Exercise 2.9

Let $0 \longrightarrow M_0 \xrightarrow{f} M_1 \xrightarrow{g} M_2 \longrightarrow 0$ be a exacte sequence of R -modules. Show that the following three statements are equivalent:

1. $M_1 \cong M_0 \oplus M_2$, with f being the inclusion of M_0 in the first summand, and g the projection onto the second.
2. f has a left-sided inverse, i. e. there is a homomorphism $h : M_1 \rightarrow M_0$ such that $h \circ f = \text{Id}_{M_0}$.
3. g has a right-sided inverse, i. e. there is a homomorphism $l : M_2 \rightarrow M_1$ such that $g \circ l = \text{Id}_{M_2}$.

Exercise 2.10

Let I, J be ideals of a ring R . Prove that there is an exact sequence of R -modules (what are the maps?) $0 \longrightarrow I \cap J \longrightarrow I \oplus J \longrightarrow I + J \longrightarrow 0$.

Exercise 2.11

1. Let $M_0 \xrightarrow{f} M_1 \xrightarrow{g} M_2 \longrightarrow 0$ be an exact sequence of R -modules and M be an R -module. Prove that, the sequence $0 \longrightarrow \text{Hom}(M_2, M) \xrightarrow{\circ g} \text{Hom}(M_1, M) \xrightarrow{\circ f} \text{Hom}(M_0, M)$ is exact.

2. Let $0 \longrightarrow M_0 \xrightarrow{f} M_1 \xrightarrow{g} M_2$ be an exact sequence of R -modules and M be an R -module. Prove that, the sequence $0 \longrightarrow \text{Hom}(M, M_0) \xrightarrow{f \circ} \text{Hom}(M, M_1) \xrightarrow{g \circ} \text{Hom}(M, M_2)$ is exact.

Chapter 3

Localization

3.1 Localization of rings

Let R be a ring. A subset S of R is called a multiplicatively closed subset if: $1 \in S$ and $\forall s, s' \in S, ss' \in S$.

Let R be a ring and S a multiplicatively closed subset of R . Define a relation $\sim$ on the set $R \times S$ by $(x, s) \sim (y, t)$ if and only if there exists $u \in S$ such that $u(tx - sy) = 0$. It is easy to see that the relation $\sim$ is an equivalent relation.

Define $S^{-1}R$ as the set of equivalence classes, and denote $\frac{x}{s}$ the class of the element (x, s) that is $\frac{x}{s} = \overline{(x, s)}$. We endowed $S^{-1}R$ with a product and sum as follows;

$$\frac{x}{s} \times \frac{y}{t} := \frac{xy}{st}, \quad \frac{x}{s} + \frac{y}{t} := \frac{tx + sy}{st}$$

The product and sum are well defined. It is easy to check that $S^{-1}R$ is a ring with $\frac{1}{1}$ for 1 and $\frac{0}{1}$ for 0. It is called the ring of fractions with respect to S or the localization at S . Note that if $s \in S$, then $\frac{s}{1}$ is an invertible element of $S^{-1}R$ with inverse $\frac{1}{s}$, since $\frac{s}{1} \frac{1}{s} = 1$.

The map $\varphi : R \rightarrow S^{-1}R$ given by $\varphi(a) = \frac{a}{1}$ is a morphism of rings. Moreover,

1. φ injective if and only if $S \cap Z(R) \neq \emptyset$.
2. $S^{-1}R$ trivial if and only if $0 \in S$.

In particular, if R is an integral domain then φ is injective.

If $S \subseteq R^\times$, then $S^{-1}R$ and R are isomorphic.

Let R be an integral domain and $S = A^*$, then S is a multiplicatively closed subset of R and $S^{-1}R$ is the fraction field of R that is $\text{Fr}(R) = S^{-1}R$.

Proposition 1.1.

Let S be a multiplicatively closed subset of R . Let $f : R \rightarrow R'$ be a morphism of rings such that, for all $s \in S$, $f(s) \in R'^{\times}$. Then there exists a unique morphism of rings $g : S^{-1}R \rightarrow R'$ such that $g \circ \varphi = f$, where $\varphi(a) = \frac{a}{1}$.

Proof :

Proposition 1.2.

Let R be a ring and S be a multiplicatively closed subset of R .

1. If I is an ideal of R , then $S^{-1}I := \{\frac{a}{s} \mid a \in I, s \in S\}$ is an ideal of $S^{-1}R$. Moreover $S^{-1}I = S^{-1}R$ if and only if $S \cap I \neq \emptyset$.
2. The ideals of $S^{-1}R$ are of the form $S^{-1}I$, where I is an ideal of R .
3. $\text{Spec}(S^{-1}R) = \{S^{-1}P \mid P \in \text{Spec}(R), P \cap S = \emptyset\}$.

Proof : ‘

Definition 1.3.

Let R be a ring.

1. Let $a \in R$ and set $S = \{a^n \mid n \in \mathbb{N}\}$. Then S is a multiplicatively closed subset of R , the corresponding localization $S^{-1}R$ is often written as R_a , we call it the localization of R at the element a .
2. Let P be a prime ideal of R . Then $S = R \setminus P$ is a multiplicatively closed subset of R , the resulting localization $S^{-1}R$ is usually denoted by R_P and called the localization of R at the prime ideal P .

Note that $R_a = \{\frac{b}{a^n} \mid a \in R, n \in \mathbb{N}\}$ and $R_P = \{\frac{a}{s} \mid a \in R, s \notin P\}$.

Remark : Let R be a ring.

1. If $a \in R$, then $\text{Spec}(R_a) = \dots\dots\dots$
2. If P is a prime ideal of R , then $\text{Spec}(R_P) = \dots\dots\dots$

Definition 1.4.

A ring R is called *local* if it has exactly one maximal ideal.

Example : $\mathbb{R}[X]/(X^2)$ is a local ring, its maximal ideal is $(\bar{X})$.

Proposition 1.5.

Let R be a ring. The following statements are equivalent;

1. R is local,
2. $R \setminus R^\times$ is an ideal.

Proof :

Proposition 1.6.

Let R be a ring and P be a prime ideal of R . Then R_P is a local ring with maximal ideal $PR_P = \{\frac{a}{s} \mid a \in P, s \notin P\}$.

Proof :

3.2 Localization of modules

Let R be a ring, S be a multiplicatively closed subset of R and M be an R -module. Define a relation on $M \times S$ by $(m, s) \sim (m', t)$ if and only if there exists $u \in S$ such that $u(tm - sm') = 0$. This relation is an equivalent relation.

Define $S^{-1}M$ as the set of equivalence classes, and denote $\frac{m}{s}$ the class of (m, s) that is $\frac{m}{s} = \overline{(m, s)}$. We endowed $S^{-1}M$ with a sum and scalar product as follows;

$$\frac{a}{s} \times \frac{m}{t} := \frac{am}{st}, \quad \frac{m}{s} + \frac{m'}{t} := \frac{tm + sm'}{st}$$

The scalar product and sum are well defined. It is easy to see that $S^{-1}M$ is an $S^{-1}R$ -module. It is called the module of fractions with respect to S or the localization at S .

The map $\varphi : M \rightarrow S^{-1}M$ given by $\varphi(m) = \frac{m}{1}$ is a morphism of R -modules.

In the cases when S is equal to $\{a^n \mid n \in \mathbb{N}\}$ for an element $a \in R$ or $R \setminus P$ for a prime ideal P of R , we will write $S^{-1}M$ also as M_a or M_P , respectively.

Proposition 2.1.

Let S be a multiplicatively closed subset of a ring R . Let N be a submodule of a module M .

1. $S^{-1}N$ is an $S^{-1}R$ -submodule of $S^{-1}M$.
2. $S^{-1}(M/N)$ is isomorphic to $S^{-1}M/S^{-1}N$.

Proof :

Theorem 2.2.

Let M be an R -module. The following statements are equivalent;

1. $M = 0$,
2. $\forall P \in \text{Spec}(R), M_P = 0$.
3. $\forall P \in \text{Max}(R); M_P = 0$.

Proof :

Let S be a multiplicatively closed subset of a ring R and $f : N \rightarrow M$ be a morphism of R -modules, then f induce a morphism of $S^{-1}R$ -modules $f_S : S^{-1}N \rightarrow S^{-1}M$ with $f_S(\frac{m}{s}) = \frac{f(m)}{s}$.

Proposition 2.3.

Let S be a multiplicatively closed subset of R and $0 \longrightarrow N \longrightarrow M \longrightarrow M' \longrightarrow 0$ be an exact sequence of R -modules. Then the following sequence $0 \longrightarrow S^{-1}N \longrightarrow S^{-1}M \longrightarrow S^{-1}M' \longrightarrow 0$ is exact.

Proof :

Theorem 2.4.

Let S be a multiplicatively closed subset of R and $f : N \rightarrow M$ be a morphism of R -modules. The following statements are equivalent;

1. $f : N \rightarrow M$ is an isomorphism.
2. $\forall P \in \text{Spec}(R), f_P : N_P \rightarrow M_P$ is an isomorphism.
3. $\forall P \in \text{Max}(R), f_P : N_P \rightarrow M_P$ is an isomorphism.

Proof :